K.R. MANGALAM UNIVERSITY

Database Management Lab File

(ENCS205)

Bachelor of Technology

in

Computer Science and Engineering

B. Tech CSE (Core)

4 Semester

School of Engineering & Technology

K. R. MANGALAM UNIVERSITY Sohna, Haryana 122103, India Jan-2026

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Experiment No. – 3

Database Schema for a customer-sale scenario

Customer(Cust id : integer, cust_name: string)

Item(item_id: integer, item_name: string, price: integer)

Sale(bill_no:integer,bill_data:date,cust_id: integer, item_id: integer, qty_sold: integer)

For the above schema, perform the following—

1. Create the tables with the appropriate integrity constraints.
2. Insert around 10 records in each of the tables.
3. List all the bills for the current date with the customer names and item numbers.
4. List the total Bill details with the quantity sold, price of the item and the final amount.
5. List the details of the customer who have bought a product which has a price>200.
6. Give a count of how many products have been bought by each customer
7. Give a list of products bought by a customer having cust_id as 5.
8. List the item details which are sold as of today.
9. Create a view which lists out the bill_no, bill_date, cust_id, item_id, price, qty_sold, amount.
10. Create a view which lists the daily sales date wise for the last one week

Code ->

```
CREATE DATABASE CustomerSaleDB;
USE CustomerSaleDB;

CREATE TABLE Customer (
   cust_id INT PRIMARY KEY,
   cust_name VARCHAR(100) NOT NULL
);

CREATE TABLE Item (
   item_id INT PRIMARY KEY,
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  item_name VARCHAR(100) NOT NULL,
  price INT NOT NULL CHECK (price > 0)
);

CREATE TABLE Sale (
  bill_no INT PRIMARY KEY,
  bill_date DATE NOT NULL,
  cust_id INT,
  item_id INT,
  qty_sold INT CHECK (qty_sold > 0),

  FOREIGN KEY (cust_id) REFERENCES Customer(cust_id),
  FOREIGN KEY (item_id) REFERENCES Item(item_id)
);

INSERT INTO Customer VALUES
(1,'Deepak'),
(2,'Sahdev Sharma'),
(3,'Kushagra'),
(4,'Aman'),
(5,'Muskan'),
(6,'Priyanka'),
(7,'Bhavik'),
(8,'Nandani'),
(9,'Vikas'),
(10,'Harshit Singh');

INSERT INTO Item VALUES
(101,'Pen',20),
(102,'Notebook',50),
(103,'Bag',500),
```

```
(104,'Shoes',1200),
(105,'Bottle',150),
(106,'Watch',800),
(107,'Keyboard',700),
(108,'Mouse',300),
(109,'Book',250),
(110,'Charger',400);

INSERT INTO Sale VALUES
(1001,CURDATE(),1,103,2),
(1002,CURDATE(),2,104,1),
(1003,CURDATE(),5,109,3),
(1004,'2026-02-20',3,101,5),
(1005,'2026-02-21',4,106,1),
(1006,'2026-02-22',5,107,2),
(1007,CURDATE(),6,108,1),
(1008,CURDATE(),7,110,4),
(1009,'2026-02-18',8,105,2),
(1010,CURDATE(),5,104,1);

SELECT S.bill_no, C.cust_name, S.item_id
FROM Sale S
JOIN Customer C ON S.cust_id = C.cust_id
WHERE S.bill_date = CURDATE();

SELECT DISTINCT C.*
FROM Customer C
JOIN Sale S ON C.cust_id = S.cust_id
JOIN Item I ON S.item_id = I.item_id
WHERE I.price > 200;
```

```
SELECT C.cust_name, COUNT(S.item_id) AS Total_Products
FROM Customer C
JOIN Sale S ON C.cust_id = S.cust_id
GROUP BY C.cust_name;

SELECT I.item_name
FROM Sale S
JOIN Item I ON S.item_id = I.item_id
WHERE S.cust_id = 5;

SELECT I.*
FROM Sale S
JOIN Item I ON S.item_id = I.item_id
WHERE S.bill_date = CURDATE();

CREATE VIEW Bill_View AS
SELECT S.bill_no, S.bill_date, S.cust_id,
    S.item_id, I.price, S.qty_sold,
    (I.price * S.qty_sold) AS Amount
FROM Sale S
JOIN Item I ON S.item_id = I.item_id;

SELECT * FROM Bill_View;

CREATE VIEW Weekly_Sales AS
SELECT bill_date,
    SUM(qty_sold * I.price) AS Daily_Total
FROM Sale S
JOIN Item I ON S.item_id = I.item_id
WHERE bill_date >= CURDATE() - INTERVAL 7 DAY
GROUP BY bill_date;
```

SELECT * FROM Weekly_Sales;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

bill_no	cust_name	item_id
1001	Deepak	103
1002	Sahdev Sharma	104
1003	Muskan	109
1007	Priyanka	108
1008	Bhavik	110
1010	Muskan	104

Result 1 × | Result 2 | Result 3 | Result 4 | Result 5 | Bill_View 6 | Weekly_Sales 7 | Read Only

cust_id	cust_name
1	Deepak
2	Sahdev Sharma
5	Muskan
4	Aman
6	Priyanka
7	Bhavik

Result 1 | Result 2 × | Result 3 | Result 4 | Result 5 | Bill_View 6 | Weekly_Sales 7 | Read Only

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

cust_name	Total_Products
Deepak	1
Sahdev Sharma	1
Kushagra	1
Aman	1
Muskan	3
Priyanka	1
Bhavik	1
Nandani	1

Result 1 | Result 2 | Result 3 × | Result 4 | Result 5 | Bill_View 6 | Weekly_Sales 7 | Read Only

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

item_name
Book
Keyboard
Shoes

Result 1 | Result 2 | Result 3 | Result 4 × | Result 5 | Bill_View 6 | Weekly_Sales 7 | Read Only

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

item_id	item_name	price
103	Bag	500
104	Shoes	1200
109	Book	250
108	Mouse	300
110	Charger	400
104	Shoes	1200

Result Grid | Form Editor | Field Types

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Bill_View 6 | Weekly_Sales 7 | Read Only

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

bill_no	bill_date	cust_id	item_id	price	qty_sold	Amount
1001	2026-02-28	1	103	500	2	1000
1002	2026-02-28	2	104	1200	1	1200
1003	2026-02-28	5	109	250	3	750
1004	2026-02-20	3	101	20	5	100
1005	2026-02-21	4	106	800	1	800
1006	2026-02-22	5	107	700	2	1400
1007	2026-02-28	6	108	300	1	300
1008	2026-02-28	7	110	400	4	1600
1009	2026-02-18	8	105	150	2	300
1010	2026-02-28	5	104	1200	1	1200

Result Grid | Form Editor | Field Types

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Bill_View 6 | Weekly_Sales 7 | Read Only

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

bill_date	Daily_Total
2026-02-28	6050
2026-02-21	800
2026-02-22	1400

Result Grid | Form Editor | Field Types

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Bill_View 6 | Weekly_Sales 7 | Read Only